

PROFESSIONAL EDUCATION

SUBCATEGORIES OF TEACHER MOVEMENT/MOVEMENT MANAGEMENT

- 1. **THRUST** – proceeding without assessing
- 2. **DANGLING** – hanging activity by giving another
- 3. **TRUNCATION** – leaves activity
- 4. **FLIP-FLOP** – returns to a left activity while currently doing an activity
- 5. **STIMULUS-BOUND** – distracted
- 6. **OVERDWELLING** – overtime in one topic
- 7. **OVERLAPPING** – multitasking results negatively

ISM's IN EDUCATION

- BEHAVIORISM** – change
- EXISTENTIALISM** – choice
- IDEALISM** – enough in mind
- PRAGMATISM** - practice (T&E)
- REALISM** – enough to see
- SOCIAL RECONSTRUCTIVISM** – benefit of all
- ESSENTIALISM** – basic
- HUMANISM** – build
- PERRENIALISM** – constant
- PROGRESSIVISM** – improve
- UTILITARIANISM** - best

AIMS OF ERAS

- PRE-SPANISH** – survival and conformity
- SPANISH** – Christianity
- AMERICAN** – democratic ideals and way of life
- COMMONWEALTH** – moral character, efficiency
- JAPANESE** – progress

PROF. ED PROPONENTS

- B.F. SKINNER** – Operant Conditioning
- BANDURA** – Modeling
- BANDURA & WALLACE** – Social Learning
- CARL JUNG** – Psychological
- CARL JUNG** – Psychological
- CONFICIUS** – Education for all, Golden Rule
- EDWARD THORNDIKE** – Connectionism
- ERICK ERIKSON** – Psychosocial
- IVAN PAVLOV** – Classical Conditioning
- JEAN PIAGET** – Cognitive
- PEZTALLOZI** – realia, Froebel's protégé
- JEROME BRUNER** – Instrumental Conceptualism
- JOHN DEWEY** – learning by doing
- JOHN LOCKE** – *Tabula Rasa* (blank sheet)
- KOHLERS** – Insight Learning
- LAURENCE KOHLBERG** – Moral Development
- LEV VGOTSKY** – Social Cognitivist, Scaffolding
- SIGMUND FREUD** – Psychosexual
- WILLIAM SHELDON** – Physiological
- FROEBEL** - Father of Kndrgrtn

PRINCIPLES

- HEDONISM** – pleasure principle
- DOUBLE EFFECT** – sacrifice for the good or bad
- FORMAL COOPERATION** – cooperation with will
- LESSER EVIL** – choice of the less one from two bad things
- MATERIAL COOPERATION** – cooperation without will

FREUD'S PSYCHOSEXUAL/PSYCHOANALYTICAL THEORY

- 1. **ORAL** (0-1 yrs. old) – Infant
- 2. **ANAL** (1-3 yrs. old) – Toddler
- 3. **PHALLIC** – Preschool
- 4. **LATENCY** – School Age
- 5. **GENITAL** – Adolescence
- OEDIPUS** – son to mom
- ELECTRA** – daughter to dad

LAWS IN EDUCATION

- PRC BR 435** – Code of Ethics for Professional Teachers
- PD 1006** – Decree Professionalizing Teachers
- RA NO. 1425** – inclusion of the works of Jose Rizal
- RA NO. 4670** – “Magna Carta for Public School Teacher”
- RA 7722** – CHED
- RA 7796** – “TESDA Act of 1994”
- RA 7836** – Phil. Teachers Professionalization Act of 1994
- RA 9155** – BEGA (Basic Educ.) or DepEd Law
- RA 9293** – Teachers Professionalization Act
- RA 10533** – K-12 Law
- ACT NO. 2706** – “Private School Law”
- COMMONWEALTH ACT NO. 578** – “persons in authority”
- KAUTUSANG PANGKAGAWARAN BLG 7** - PILIPINO NatlLng
- PROKLAMA BLG 12** - Linggo ng Wika (Balagtas,Mr29-Ap4)
- PROKLAMA BLG. 186** – Linggo ng Wika (Quezon,Ag13-19)
- PROKLAMA BLG. 1041** – Buwan ng Wika (Ramos)
- PHIL. CONSTITUTION ACT 14** – ESTACS
- RA 1079** – no limit of Civil Service eligibility
- RA 6655** – “Free Public Secondary Educ. Act of 1988”
- RA 6728** – “Act Providing Government Assistance to Students and Teachers in Private Education
- RA 7277** – Magna Carta for PWD
- RA 7610** – Anti-Child Abuse Law (Amendment: RA 9231)
- RA 7743** – establishment of public libraries
- RA 7877** – “Anti Sexual Harassment Act of 1995”
- RA 7880** – “Fair and Equitable Access to Education Act”
- RA 8049** – Anti-Hazing Law
- RA 8187** – Paternity Act
- RA 10627** – Anti-Bullying
- SB 1987 ART. 14 SEK. 6-9** – FILIPINO (National Language)

BRUNER'S THREE MODES OF REPRESENTATION

- 1. **ENACTIVE** (0-1 yrs. old) – action-based information
- 2. **ICONIC** (1-6 yrs. old) – image-based information
- 3. **SYMBOLIC** (7+) – code/symbols such as language

TAXONOMY OF OBJECTIVES

COGNITIVE:

- | BLOOM (LOTS) | ANDERSON (HOTS) |
|-----------------|-----------------|
| o Knowledge | o Remembering |
| o Comprehension | o Understanding |
| o Application | o Applying |
| o Analysis | o Analyzing |
| o Synthesis | o Evaluating |
| o Evaluation | o Creating |

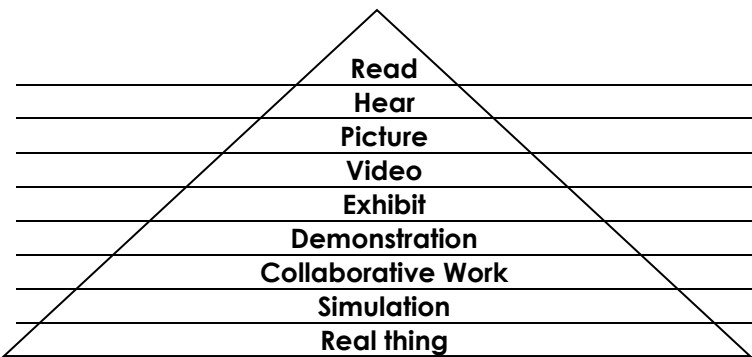
AFFECTIVE:

- o Receiving
- o Responding
- o Valuing
- o Organizing
- o Characterization

PSYCHOMOTOR:

- | SIMPSON | HARROW |
|--------------------------|--------------------------------|
| o Perception | o Reflex movement |
| o Set | o Fundamental Movement |
| o Guided Response | o Physical Movement |
| o Mechanism | o Perceptual Abilities |
| o Complex Overt Response | o Skilled Movements |
| o Adaptation | o Non-discursive communication |
| o Origination | |

DALES CONE OF EXPERIENCE



ERIKSON'S PSYCHOSOCIAL TASKS

1. **TRUST VS. MISTRUST** (0-12 months)
2. **AUTONOMY VS. SHAME/DOUBT** (1-3 years old)
3. **INITIATIVE VS. GUILT** (3-6 years old)
4. **INDUSTRY VS. INFERIORITY** (6-12 years old)
5. **IDENTITY VS. ROLE CONFUSION** (12-18 years old)
6. **INTIMACY VS. ISOLATION** (early 20s-early 40s)
7. **GENERATIVITY VS. STAGNATION** (40s-mid 60s)
8. **INTEGRITY VS. DESPAIR** (mid 60s-death)

PIAGET'S COGNITIVE DEVELOPMENT THEORY

1. **SENSORY** – senses
2. **PRE-OPERATIONAL** – imagination
3. **CONCRETE**
4. **FORMAL**

GENERATIONS OF COMPUTER

1. **VACUUM TUBES** (1940-1956)
2. **TRANSISTORS** (1956-1963)
3. **INTEGRATED CIRCUITS** (1964-1971)
4. **MICROPROCESSORS** (1971-present)
5. **ARTIFICIAL INTELLIGENCE** (present-future)

MISTAKEN GOALS

1. **ATTENTION SEEKER** – “teacher, notice me”
2. **REVENGE** – “teacher, I am hurt”
3. **POWER-SEEKING** – “teacher, may I help?”
4. **INADEQUACY** – “teacher, don't give up on me”
5. **WITHDRAWAL** – “teacher, please help me”

KOHLBERG'S THEORY OF MORAL DEVELOPMENT

LEVEL 1: PRE-CONVENTIONAL MORALITY

Stage 1: Obedience and Punishment Orientation

Stage 2: Individualism and Exchange

LEVEL 2: CONVENTIONAL MORALITY

Stage 3: Good Interpersonal Relationships

Stage 4: Maintaining the Social Order

LEVEL 3: POST-CONVENTIONAL MORALITY

Stage 5: Social Contract and Individual Rights

Stage 6: Universal Principles

CENTRAL TENDENCY -Central (middle location) Tendency

MEAN – Average

MODE – most occurring

RANGE – highest score minus lowest score

LOW SD–Homogenous, scores near to mean(almost same)

HIGH SD – Heterogenous, scores far to mean (scattered)

DECILE – 10 grps (D1...D10) **QUARTILE** – 4 grps (Q1...Q4)

SUSPENSION – time

REVOKATION – condition

DIFFICULTY INDEX

0-0.20	VERY DIFFICULT
0.21-0.40	DIFFICULT
0.41-0.60	MODERATELY DIFFICULT
0.61-0.80	EASY
0.81-1.00	VERY EASY

POSITIVELY SKEWED (*LEFT FOOT*)

- low scores, mean greater than mode

NEGATIVELY SKEWED (*RIGHT FOOT*)

- high scores, mean is lower than mode

HORN/HALO EFFECT

- overcoming other trait, either bad/good

GENERAL EDUCATION

FILIPINO/ENGLISH:

MGA TEORYA NG PINAGMULAN NG WIKA

1. **BOW-WOW** –kalikasan at hayop
2. **DING-DONG** – bagay
3. **POOH-POOH** – masidhing damdamin
4. **YOHEHO** – pwersang pisikal

FILIPINO POETS AND PEN NAMES

BENVENIDO SANTOS – American Culture Writings

DANIEL DEFOE – “Robinson Crusoe” (novel)

EDILBERTO TIEMPO – made “Cry Slaughter” that has been translated many times

ERNEST HEMINGWAY – Ring Lardner Jr.

JOSE GARCIA VILLA – “Comma Poet”, Dove G. Lion

MIGUEL DE CERVANTES/ESCALANDE - Masterpiece is “Don Quixote” that is most influential

NICK JOAQUIN – Quijano de Manila
– Spanish Culture Writing

PAZ MARQUEZ BENITEZ – made “Dead Stars” that is the 1st modern English short story

SAMUEL LANGHORNE CLEMENS – Mark Twain

- “Adventures of Tom Sawyer”

- “Adventures of Huckleberry Finn” (best novel)

SINTAKS/SINTAKSIS/PALAU GNAYAN

- sangay ng barirala na tumatalakay sa masistemang pagkaka-ayos-ayos ng mga salita sa pagbuo ng mga parirala at pangungusap

PEN NAMES OF FILIPINO PROTAGONISTS

ANDRES BONIFACIO – May Pag-asa, Agapito Bagumbayan

ANTONIO LUNA – Taga-Ilog

EMILIO AGUINALDO - Magdalo

EMILIO JACINTO – Di Masilaw, Tingkian

GRACIANO LOPEZ JAENA – Diego Laura

JOSE MA. PANGANIBAN – JoMaPa

JOSE RIZAL – Dimas alang, Laon laan

JUAN LUNA – Buan

MARCELO DEL PILAR – Plaridel, Dolores Manapat, Piping Dilat

MARIANO PONCE – Tikbalang, Naning (Satanas), Kalipulako

MGA URI NG PANGUNGUSAP

WALANG PAKSA:

1. **EKSISTENSYAL** – mayroong isa o higit pang tao
Halimbawa: Mayroon daw puno sa bakuran.
2. **MODAL** – nais/pwede/maari (*Gusto ko matulog.*)
3. **PANLIPUNAN** – pagbati, pagbigay galang atbp.
4. **SAGOT LAMANG** – “Talaga?”, “Oo”
5. **SAMBITLA** – masidhing damdamin (*Aray!*)
6. **TEMPORAL** – panandaliang kalagayan o panahon

KAYARIAN:

1. **PAYAK** – iisang kaisipan
2. **TAMBALAN** – dalawang sugnay na ‘di makapag-iisa
3. **HUGNAYAN** – madalas nagsisimula sa *kung, dahil sa*
4. **LANGKAPAN** – mahabang pangungusap

MGA URI NG KWENTO

1. **PABULA** (*fable*) – hayop
2. **PARABULA** (*parable*)– Bibliya
3. **ANEKDOTA** (*anecdote*) – tunay na buhay
4. **MITOLOHIYA** (*myth*) – diyos at diyosa (pinagmulan)

ASPEKTO NG PANDIWA (Verb)

1. **PERPEKTIBO** – tumakbo
2. **IMPERPEKTIBO** – tumatakbo
3. **KONTEMPLATIBO** – tatakbo

KAANTASAN NG PANG-URI (Adjective)

1. **LANTAY** – walang pinaghahambing
2. **PAHAMBING** – inihalalintulad
3. **PASUKDOL** – nangingibabaw (H: pinakamataas)

MGA URI NG TULA

1. **PATULA** (Moro-moro)
2. **PASALAYSAY** (Epiko, Awit, Korido)

MGA AWITING BAYAN

1. **DALIT/HIMNO** – pagsamba sa anito o pang-relihiyon
2. **DIONA** – kasal
3. **DUNG-AW** – patay (pagdadalamhati)
4. **KALUSAN** – paggawa
5. **KUMINTANG** – tagumpay (pandigma)
6. **KUNDIMAN** – pag-ibig
7. **OYAYI** – pagpapatulog ng bata
8. **SOLIRANIN** – pagsasagwan
9. **TALINDAW** – pamamangka

PAGBABAGONG MORPONEMIKO

1. **ASIMILASYON** – Parsyal (pangsukli), Ganap (panukli)
2. **MAY ANGKOP** – wikain mo – “kamo”
3. **MAYSUDLONG/PAGDARAGDAG NG PONEMA**
- muntik – muntikan, pagmuntikan, pagmuntikanan
4. **METATESIS** – linipad – nilipad
5. **PAGKAKALTAS NG PONEMO** – takipan – takpan
6. **PAGLILIPAT-DIIN** – laRUan (playground) - laruAN (toy)
7. **PAGPAPALIT NG PONEMA** – madapat – marapat

MGA URI NG PANGHALIP/PRONOUNS

1. **PANAO/PERSONAL PRONOUN** – ako/I etc.
2. **PAMATLIG/DEMONSTRATIVE PRONOUN** – ito/this etc.
3. **PANAKLAW/INDEFINITE P.** - isa, all, anyone etc.
4. **PATULAD** – ganito, ganyan atbp.
5. **PANANONG/INTERROGATIVE P.** – sino, when etc.
6. **PAMANGGIT/RELATIVE P.** – daw, umano, which, who

MGA AKDANG NA MAY IMPUWENSYA SA MUNDO

AKLAT NG MGA ARAW – China (by Confucius)
AKLAT NG MGA PATAY – Egypt cults & myths (by Osiris)
AWIT NI ROLANDO-France (by Doce Pares, Roncesvalles)
BIBLIYA – Palestino at Greece
CANTEBURY TALES – America (by Chaucer)
DIVINE COMEDIA – Italy (by Dante)
EL CID COMPEADOR – katangian at history ng Spain
ILIAD o ODYSSEY – Myths of Greece made by Homer.
ISANG LIBO'T ISANG GABI – Ugali sa Arabia at Persia
KORAN – Arabia (Muslim Bible) **MAHABRATA** – India
UNCLE TOM'S CABIN – about slaves that becomes the basis of democracy. (by Harriet Beecher Stowe of U.S.)

MGA URI NG PANITIKAN

1. **TULUYAN** – binubuo ng mga pangungusap
 - A. **NOBELA** – binubuo ng mga kabanata
 - B. **DULA** – pagtatanghal sa entablado**MGA DULANG PANLIBANGAN:**
 - a. **TIBAG** – Sta. Elena
 - b. **LAGAY** – Pilarenos ng Sorsogon
 - c. **PANUNULUYAN** – pagtatanghal bago mag-alas dose (12PM) ng gabi ng kapaskuhan
 - d. **PANUBOL** – parangal sa may kaarawan
 - e. **KARILYO** – ala-puppet show
 - f. **KURIDO** – katapangan, kabayanihan, kababalaghan, pananampalataya
 - g. **SARSUELA** – musical tungkol sa pag-ibig, paghihiganti atbp. masisidhing damdamin
- C. **ALAMAT** – pinagmulan
- D. **ANEKDOTA** – ugali, may mabuting aral
2. **PATULA** - may sukat, pantig, tugma, taludtod, saknong
 - A. **TULANG PASALAYSAY** - mahahalagang tago o pangyayari sa buhay.
 - a. **EPIKO** – kabayanihan sa kababalaghan
 - o **BIDASARI, PARANG SABIR** – Moro
 - o **BIAG NI LAM ANG** – Iloko
 - o **MARAGTAS, HARAYA, LAGDA AT HARI SA BUKID** – Bisaya
 - o **KUMINTANG** – Tagalog
 - o **DAGOY AT SUDSUD** – Tagbanua
 - o **TATUANG** - Bagobo
 - b. **AWIT o KORIDO** - kaharian
 - c. **TULA NG DAMDAMIN o LIRIKI** – own feeling**MGA TULANG LIRIKO:**
 - o **AWITING BAYAN** – kalungkutan
 - o **ELEHIYA** – yumao
 - o **DALIT** – pagpupuri sa Diyos
 - o **PASTORAL** – buhay sa bukid
 - o **ODA** – papuri
 - B. **TULANG DULA O PANGTANGHALAN**
 - a. **KOMEDYA**
 - b. **MELODRAMA** – musical
 - c. **TRAHEDYA** – death of main character
 - d. **PARSA** – mga pangyayaring nakakatawa
 - e. **SAYNETE**
-karaniwang pag-uugali ng tao/ pook

C. TULANG PATNIGAN

- a. **KARAGATAN** – alamat ng singsing ng prinsesa na naihulog niya sa dagat sa hangaring mapangasawa ang kasintahang mahirap.
- b. **DUPLO** – paligsahan ng husay sa pagtula
- c. **BALAGTASAN** – pumalit sa Duplo

FIGURES OF SPEECH/TAYUTAY

PAG-UUGNAY O PAGHAHAMBING:

1. **SIMILE/PAGTUTULAD** – mayroong pangatnig
2. **METAPHOR/PAGWAWANGIS** – walang pangatnig
3. **ALUSYON** – iba't ibang aspekto ng buhay ng tao
4. **METONYMY/PAGPAPALIT-TAWAG**
5. **SYNECDOCHE/SINEKDOKE** – pagbanggit ng isa upang tukuyin ang kabuuan
Hal: Dalawang bibig ang umaasa kay Romeo.

PAGLALARAWAN:

6. **HYPERBOLE/ PAGMAMALABIS o EKSAHERASYON**
7. **APOSTROPHE/PAGTAWAG** – pakikipag-usap sa hindi buhay o malayong tao. Hal: Ulan, tumigil ka na.
8. **EXCLAMATION/PAGDARAMDAM** – strong feeling.
9. **PARADOX/PARADOKS** – “malayo ma'y malapit pa rin”
10. **OXYMORON/PAGTATAMBIS** - paradox w/ extra words

PAGSASALIN NG KATANGIAN:

PERSONIFICATION/PAGSASATAO

PAGSASATUNOG:

12. **ONOMATOPOEIA/PANGHIHIMIG** – tunog ang paksa
13. **ALLITERATION/PAG-UULIT** – repetition of 1st letter in the 1st word. Ex: Dinggin mo ang Diyos na Dinadakila
14. **REPITASYON** – repetition of phrase. Ex: Tama! Tama!...

IBA PANG TAYUTAY NA GAMIT SA TULA:

ALITERASYON – unang titik o pantig ay pare-pareho
ANADIPLOSIS – paggamit ng salita sa unahan at hulihan
EPIORA – pag-uulit ng salita sa hulihan
PAG-UYAM – sarcasm
LITOTES – pagtanggap o pagkukunwari.

TALUDTOD – linya sa tula

UNFAMILIAR PARTS OF THE SPEECH

1. **PREPOSITIONS** – on, under, off, by, in near, for, to, since
2. **CONJUNCTIONS (PANGATNIG)**
- para/for, at/and, nor, or, pero/but, yet, so, ni, ngunit
3. **INTERJECTION** – with exclamation mark

PROPER SEQUENCE OF WORDS IN A SENTENCE

1. **ARTICLES** – a, an, the
2. **OPINION**
3. **SIZE**
4. **AGE**
5. **SHAPE**
6. **COLOR**
7. **MATERIAL**
8. **PURPOSE**

CLASSIFICATIONS OF POEM

1. **BALLAD** – narrative, less folk tale/legend, to be sung
2. **BLACK VERSE** – with meter but no rhyme
3. **DRAMATIC MONOLOGUE** – written in form of speech for individual character.
4. **ELEGY** – death of individual
5. **EPIC** – tells a story about heroic figure
6. **EULOGY** – message for the dead
7. **FREE VERSE** (*vers libre*) – without meter but with rhyme
8. **HAIKU** – Japanese poem about nature. 5, 7, 5 (3 lines and 17 syllables)
9. **IDYLL** (Idyl) – peaceful, idealized country scene
10. **LYRICS** - thoughts and feelings
11. **NARRATIVE** – tells story
12. **ODE** -typically serious/meditative nature, type of Lyric
13. **PASTORAL** –rural life in peaceful & romanticized way
14. **SONNET** – Lyric poem consists of 14 lines
15. **TANKA** – Japanese poem: 5 lines, 31 syllables

PHILIPPINE CONSTITUTIONS

BIAK NA BATO – pact, thought of 1st Republic.

- Spainards paid P200 000

1. **MALOLOS CONSTITUTION** – Apolinario Mabini
 - rights of soldiers
 - no Visayas yet in right of territories
2. **1935 CONSTITUTION** – adapted from American Const.
3. **1943 CONSITUTION** - Jose P. Laurel
 - Japan invades but gave freedom for Phil. to rule.
4. **1937 CONSTITUTION** – Ferdinand Marcos
 - Martial Law – 60days max
 - Nat'l Territory forced Kalayaan grp. of Islands & Saba
5. **1987 CONSTITUTION** - 18 articles
 - past chairwoman: Cecilia Muñoz Palma (Feb 2, 1987)
 - Bill of Rights are for the criminals

JUS SANGUINI – blood

JUS SOLI – place

SOME TYPES OF GOVERNMENTS

1. **COMMUNIST** – classless society
 - State plans and controls economy
2. **PARLIAMENTARY** – majority of people voted
3. **REPUBLICAN** – power comes from people

PHILIPPINE PRESIDENTS

(AgQueLaOsRoQuiMagGarMaMarAquiRaEsArAquiDut)

ACTS

1. **ASSOCIATION OF SE ASIAN NATIONS (ASEAN)**
 - Myanmar, Cambodia, Laos, Brunei, Philippines, Singapore, Thailand, Indonesia, Vietnam, Malaysia
2. **BELL TRADE (PHILIPPINE TRADE ACT)** – bet. Phil. & U.S.
3. **KYOTO PROTOCOL (UNNCC)**
 - fight global warming decreasing green house gases
4. **RIO DE JANEIRO CONVENTION**
 - environment and sustainable development
5. **TEJEROS CONVENTION** – election
 - Bonifacio elected as Director Imperior
6. **TREATY ON GENERAL RELATIONS**
 - recognition of U.S. to Philippine freedom
7. **UNDERWORLD-SIMMONS ACT** – full free foreign trade
8. **PAYNE ALDRICH ACT** – partial free foreign trade

MISSIONARIES AND EXPEDITIONS

1. **AUGUSTINIAN** – most intelligent
2. **FRANCISCAN** – sends medical aids
3. **JESUITS**
4. **DOMINICANS** – richest
5. **RECOLECTS** – most killed schools

MARTYR PRIESTS

1. **BURGOS** – youngest, mastermind of secularization
2. **GOMEZ** – Oldest, likes “sabong” and hid there

HOMO HABILIS – man of steel (bighead, uses muscle)

HOMO ERECTUS – man who discovered fire & clothes

HOMO SAPIENS – thinking man (can produce materials)

UNFAMILIAR BRANCHES OF BIOLOGY

ANATOMY – inner organs **ENTOMOLOGY** – insects

BIOCHEMISTRY – chemical patterns of animals

ECOLOGY – living things bet. Each other in environment

EMBRYOLOGY – developmental patterns fr. zygote-birth

GENETICS – heredity **MYCOLOGY** – fungi

HERPETOLOGY – reptiles and amphibians

HISTOLOGY – plant and animal tissues

MORPHOLOGY – phenotype (appearance)

ORNITHOLOGY – birds **PARASITOLOGY** – parasites

PALEONTOLOGY – fossils of animals and plants

PHYSIOLOGY – function of tissue, organ & system

TAXONOMY – classification of living organisms

SCIENCE PROPONENTS

CAROLUS LINNAEUS – Father of Taxonomy

ROBERT HOOKE – termed “cells” (*cellulae*)

ANTON VAN LEUWENHOEK – 1st person to observe microscopic organisms (animal cule)

ROBERT BROWN – discovered Nucleus

MATTHIAS SCHIEDEN (Botanist) & **THEODORE** (Zoologist)

- found all plants consist of cells

RUDOLF VIRCHOW – proposed cells come fr. existing cells

EARTH'S SPHERES

ATMOSPHERE – gaseous sphere protection from meteors

Divided into five:

- Troposphere
- Stratosphere
- Mesosphere
- Thermosphere
- Exosphere

HYDROSPHERE – water

LITHOSPHERE – oceanic and continental crust

BIOSPHERE – all life forms in Earth

CRYOSPHERE – ice

ANTHROSPHERE – ancestors

PLANETS AND THEIR SEQUENCE

1. **SUN** – 99.86% of Solar System
 - believed was formed 4.6 billion years ago
 - Responsible for weather and climate
2. **MERCURY** – named after Roman God
 - no satellite and atmosphere
 - discovered by Mariner Ten
3. **VENUS** – Goddess of Love and Beauty (Mariner 2)
 - Perfect sphere, sister planet of Earth
4. **EARTH**
5. **MARS** – God of War, red planet (Mariner 9)
6. **JUPITER** -Gas Giant, fastest rotating planet (10hrs less)
 - has Great Red Spot: huge storm for 350yrs
7. **SATURN** – God of Agriculture (chunks of rocks)
 - made mostly of hydrogen
8. **URANUS** – Frederick William He